

ORIGINAL RESEARCH

Metabolites Associated With Long-Term Risk of Peripheral Artery Disease: The ARIC Study

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BACKGROUND: Lower extremity peripheral artery disease (PAD) is a major contributor to cardiovascular morbidity and mortality; however, the metabolic pathways underlying disease development and progression are poorly understood.

METHODS: We conducted a prospective cohort study of 3759 participants from the ARIC study (Atherosclerosis Risk in Communities; mean age, 53.5 years; 60% women; 62% Black participants) with untargeted metabolomic profiling at baseline (1987–1989). Incident PAD (458 cases) and its severe form with rest pain or tissue loss, critical limb ischemia (CLI; 135 cases), were identified through hospitalization codes over a median follow-up of 27 years. We used Cox regression with adjustment for traditional risk factors and accounted for multiple testing using a false discovery rate correction at a threshold of 0.05. Risk prediction improvement was evaluated using Harrell C statistics and the net reclassification improvement.

RESULTS: Thirteen metabolites were significantly associated with incident PAD and primarily reflected dysregulated glycemic control, impaired redox balance, and altered lipid remodeling (eg, higher levels of mannose and lower levels of 1,5-anhydroglucitol, gamma-glutamyl dipeptides, and lysophospholipids). For CLI, 18 metabolites were identified, 11 of which were unique to CLI and reflected intensified oxidative and bioenergetic stress (eg, homocitrulline, arabinonate, and 1-linoleoylglycerol). Metabolites significantly improved risk prediction for PAD beyond traditional risk factors (Δ C statistic, 0.017 [95% CI, 0.008–0.025] from a base C statistic of 0.770; net reclassification improvement, 0.053 [95% CI, 0.018–0.083]) and particularly CLI (Δ C statistic, 0.041 [95% CI, 0.018–0.064] from a base C statistic of 0.823; net reclassification improvement, 0.176 [95% CI, 0.085–0.268]).

CONCLUSIONS: Distinct metabolomic profiles were associated with incident PAD and CLI years before clinical onset, highlighting dysregulated glycemic control, impaired redox balance, and altered lipid remodeling as key underlying pathways. These findings suggest that circulating metabolites may serve as early indicators of metabolic health relevant to PAD risk identification and progression to advanced disease.

Key Words: chronic limb-threatening ischemia ■ dipeptides ■ homocitrulline ■ mannose ■ metabolic networks and pathways

Lower extremity peripheral artery disease (PAD) is a progressive atherosclerotic condition caused by occlusion of the leg arteries and encompasses a wide clinical spectrum ranging from asymptomatic disease to critical limb ischemia (CLI).¹ PAD affects >230 million individuals globally^{2,3} and contributes substantially to morbidity, including leg amputations, and mortality.^{4–6} Improving outcomes in PAD depends on early disease detection and the timely implementation of guideline-directed risk management strategies.⁷ Despite this, PAD remains underdiagnosed and undertreated, particularly in

asymptomatic individuals who account for approximately two-thirds of cases.⁸

Although insights from coronary heart disease (CHD) research are often extrapolated to PAD, longstanding evidence suggests important differences in risk factor profiles and biological pathways between these conditions. For example, smoking and diabetes exert stronger effects on PAD compared with CHD, whereas dyslipidemia plays a more important role in CHD than in PAD.^{9–11} Moreover, multiple studies have demonstrated that microvascular injury, assessed using distal perfusion measures, such as

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Nonstandard Abbreviations and Acronyms

1,5-AG	1,5-anhydroglucitol
ARIC	Atherosclerosis Risk in Communities
CHD	coronary heart disease
CLI	critical limb ischemia
CEHC	carboxyethyl hydroxychroman
HDL	high-density lipoprotein
HR	hazard ratio
PAD	peripheral artery disease

toe-brachial index decline, inflammatory and endothelial biomarkers, and indicators of oxidative stress, contributes more prominently to the development and progression of PAD, especially in its advanced form, CLI, than to CHD or stroke.^{12–16} These observations highlight the unique pathophysiology of PAD and underscore the need to investigate metabolomics specifically for PAD.

Metabolomics provides a powerful approach to uncover novel biomarkers and pathways underlying complex diseases. In cardiovascular research, metabolomic profiling has identified distinct signatures for CHD and ischemic stroke. A large meta-analysis identified 56 circulating metabolites, mainly lipids and amino acids, that were associated with incident CHD.¹⁷ Similarly, studies of ischemic stroke have implicated alterations in branched-chain amino acids, lysophospholipids, and energy metabolism, pointing to disrupted pathways such as oxidative stress and neuroinflammation.¹⁸ Among patients with clinical manifestations suggestive of stroke, a 4-metabolite panel distinguished ischemic stroke from nonstroke causes with greater accuracy than computed tomography.¹⁹ Together, these findings illustrate the potential of metabolomics to capture vascular disease-specific biology and motivate its application to PAD.

Metabolomic research in PAD remains limited. To the best of our knowledge, of the 7 published studies directly investigating metabolites in PAD,^{20–26} 5 were cross-sectional,^{20–23,26} 4 explored targeted metabolites,^{21,23,25,26} and most were small (sample size <500).^{20–23,26} Therefore, we aimed to comprehensively investigate untargeted circulating metabolites and their associations with incident PAD and its severe form, CLI, using prospective data from the ARIC study (Atherosclerosis Risk in Communities), one of the largest US population-based cohorts with 30 years of follow-up.

METHODS

Data Availability

ARIC data are available from the National Heart, Lung, and Blood Institute Biologic Specimen and Data Repository Information Coordinating Center (<https://biolincc.nhlbi.nih.gov/>

What Are the Clinical Implications?

This study identifies serum metabolites as early indicators of long-term risk of peripheral artery disease and critical limb ischemia, highlighting metabolic pathways related to glycemic control, redox balance, and lipid remodeling that may precede clinically recognizable disease. The broader and partly distinct metabolomic profile associated with critical limb ischemia suggests that limb-threatening ischemia may reflect biological processes beyond the anatomic severity of peripheral artery disease, with potential relevance for understanding disease progression and identifying individuals at higher risk. The modest improvement in risk prediction beyond traditional risk factors suggests that metabolite panels may have future utility for refining long-term peripheral artery disease and critical limb ischemia risk assessment, pending validation in independent cohorts.

studies/aric/). Researchers may also contact the ARIC study coordinating center for data access.

Study Population

The ARIC study is a prospective community-based cohort of 15792 individuals who were enrolled between 1987 and 1989 from 4 US communities (Forsyth County, NC; Jackson, MS; Minneapolis suburbs, MN; and Washington County, MD). Participants underwent serial clinic examinations and were followed longitudinally for cardiovascular events and mortality.²⁷ Institutional review boards at each center and the data coordinating center approved the study, and written informed consent was obtained from participants at baseline and follow-up visits.

In the ARIC study, a random selection of serum samples collected at visit 1 from 1977 Black participants from the Jackson site was sent for metabolomic profiling at Metabolon Inc (Durham, NC) in 2010 (batch 1). A second random selection of samples from an additional 2055 participants (including 575 Black participants) from all 4 study sites was similarly processed in 2014 (batch 2).^{28,29} Of the 4032 participants with metabolomic assessments, we excluded participants with prevalent PAD (defined as a self-reported history of leg revascularization, intermittent claudication, or ankle-brachial index ≤ 0.9 at visit 1; $n=215$) or those with missing required visit 1 or stroke-derived linkage data ($n=58$), leaving a total of 3759 participants for the present study (Figure S1). This observational cohort study was reported in accordance with the Strengthening the Reporting of Observational Studies in Epidemiology guidelines, and a completed Strengthening the Reporting of Observational Studies in Epidemiology checklist is provided in the Supplemental Material.

Assessment of Metabolites

Serum metabolites were measured in baseline fasting samples (collected after an overnight fast of at least 8 hours) that had been stored continuously at -80°C since collection. More than 600 named and unnamed metabolites were identified using an

untargeted gas chromatography–mass spectrometry and liquid chromatography–mass spectrometry approach.^{30,31} Metabolite processing and data cleaning have been described in detail in previous studies.^{32–34} Metabolites were excluded from analysis if they had >80% of measurements below the detection limit or had a variance on the log scale <0.01 or missing variance. Remaining values below the detection limit were imputed to the minimum detected value for each metabolite across all samples. The distribution of metabolite values below the detection limit is shown in Figure S2. Metabolite distributions were then log-transformed and standardized before analysis to account for skewed distributions and improve comparability across metabolites. For this study, we included 317 named endogenous compounds with a known chemical identity available in both batches (137 in lipid metabolism pathways, 87 in amino acid, 49 in peptide, 15 in nucleotide, 13 in carbohydrate, 11 in cofactors and vitamins, and 5 in energy; Figure S3). We did not include exogenous (drug) metabolites in our analysis because we were interested in internal pathophysiological pathways and metabolic perturbations involved in PAD. As previously reported, quality control was performed on 97 replicate samples in both batches with a median correlation of 0.71 across the 285 metabolites.^{28,32}

Assessment of Incident PAD and CLI

We assessed incident PAD and CLI from PAD-related hospitalizations using *International Classification of Diseases, Ninth Revision* and *Tenth Revision* codes. Hospitalizations were identified through annual (semiannually since 2012) phone interviews and active surveillance of hospital records in the 4 ARIC communities. Incident PAD was identified using the following *International Classification of Diseases, Ninth Revision* codes (up to September 30, 2015) and corresponding *International Classification of Diseases, Tenth Revision* codes (after October 1, 2015) based on previous literature^{35,36}: atherosclerosis of native arteries of the extremities, unspecified (440.20); atherosclerosis of native arteries of the extremities with intermittent claudication (440.21); atherosclerosis of native arteries of the extremities with rest pain (440.22); atherosclerosis of native arteries of the extremities with ulceration (440.23); atherosclerosis of native arteries of the extremities with gangrene (440.24); other atherosclerosis of native arteries of the extremities (440.29); atherosclerosis of bypass graft of the extremities (440.3); atherosclerosis of other specified arteries (440.8); peripheral vascular disease, unspecified (443.9); and leg artery revascularization (38.18, 39.25, 39.29, 39.50). Of these, 440.22, 440.23, and 440.24 were considered CLI. In addition, we included any PAD cases with the following codes as CLI: leg amputation (84.1x), lower extremity ulcer (707.1x), and gangrene (785.4). Participants were censored at death, loss to follow-up, or December 31, 2022, whichever occurred first.

Assessment of Covariates

Baseline covariate information was collected at visit 1 using standardized procedures, including medical history interviews, physical examinations with anthropometric measurements, and laboratory assessments. Age, sex, race, and education level were collected from interviewer-administered questionnaires. As previously described in ARIC, we created a race-center

variable to account for nonuniform racial distributions across study centers.²⁸ Smoking was self-reported and quantified in pack-years. Physical activity was measured using a sport index during leisure time from the Baecke questionnaire.³⁷ Diabetes was defined as fasting blood glucose ≥ 126 mg/dL, nonfasting blood glucose ≥ 200 mg/dL, self-reported history of physician-diagnosed diabetes, or use of diabetes medications. Prevalent cardiovascular disease was defined as a history of CHD or stroke. History of CHD was defined as a self-reported history of myocardial infarction or coronary revascularization or electrocardiogram reading of silent myocardial infarction.³⁸ Prevalent stroke was defined based on a self-reported history of physician-diagnosed stroke or transient ischemic attack.³⁹ Blood pressure was measured 3x with a random zero-mercury sphygmomanometer, and the second and third measurements were averaged.⁴⁰ Estimated glomerular filtration rate was calculated using the creatinine-based Chronic Kidney Disease Epidemiology Collaboration 2021 equation.⁴¹ Standardized enzymatic procedures adapted to the Cobas-Bio Analyzer of Roche were used for measuring total cholesterol and HDL (high-density lipoprotein) cholesterol.⁴² Medication use was defined as the use of medication to treat a condition during the previous 2 weeks and confirmed by medication containers brought to the visit by each participant.³⁸

Statistical Analysis

We compared baseline characteristics across incident PAD status (no PAD, PAD without CLI, and PAD with CLI) using χ^2 tests for categorical variables and ANOVA for continuous variables.

We used Cox proportional hazards models to identify individual metabolites associated with incident PAD and CLI as separate outcomes. We adjusted for age, sex, race-center, education level, smoking, physical activity, diabetes, systolic blood pressure, total and HDL cholesterol, prevalent cardiovascular disease, medication use (antihypertensive and lipid-lowering), estimated glomerular filtration rate, and batch. Regression models used complete-case analysis based on the covariates included in each model. For metabolites significantly associated with PAD or CLI, we assessed model diagnostics using scaled Schoenfeld residuals to assess the proportional hazards assumption, Martingale residuals to assess potential non-linearity of continuous predictors, and deviance residuals to examine potentially influential observations.⁴³ All hazard ratios (HRs) represent the risk per 1-SD increment in metabolite level. We used the Benjamini-Hochberg false discovery rate procedure with a threshold of <0.05 to account for multiple testing, corresponding to significance cutoffs of $P < 2.04 \times 10^{-3}$ and $P < 2.83 \times 10^{-3}$ for PAD and CLI, respectively. To formally assess whether significant metabolite associations differed in magnitude between CLI and PAD, we used seemingly unrelated regression models that jointly estimated effects for both outcomes while accounting for correlated errors. We defined significant disproportionate associations with CLI relative to PAD by a $P < 2.90 \times 10^{-2}$. All reported *P* values were false discovery rate-adjusted unless otherwise stated. In addition, we repeated the analyses stratified by diabetes, smoking status, and sex for significant metabolites, with interactions formally assessed using likelihood ratio tests.

For metabolites with significant associations with PAD or CLI, we assessed their predictive performance by calculating Harrell C statistic,^{44,45} a measure of risk discrimination.

We also calculated both the continuous and categorical net reclassification index based on the 30-year risk of incident PAD or CLI.⁴⁶ For the categorical net reclassification improvement, participants were stratified into low-, intermediate-, and high-risk groups based on 30-year risk of <8%, 8% to 23%, and >23% for PAD and <2%, 2% to 6%, and >6% for CLI. Because we are not aware of established risk thresholds for PAD, these thresholds were based on the 30-year average risk of both outcomes (PAD 16% and CLI 5%) in our study population. The base model was prespecified using established PAD risk factors—age, sex, smoking, systolic blood pressure, total and HDL cholesterol, and diabetes.⁴⁷ Regression coefficients for this model were estimated in our study population and used consistently as the reference model for all prediction analyses. Standard errors for all performance metrics were estimated using 1000 bootstrap replications.

We conducted several prespecified sensitivity analyses to evaluate the robustness of our primary findings. First, to assess whether associations were driven primarily by the very long-term follow-up, we repeated the Cox proportional hazards models, restricting follow-up to 15 years. These analyses were limited to metabolites identified as significant in the primary analyses for PAD (13 metabolites) and CLI (18 metabolites), and used identical covariate adjustment as in the primary models. Second, to evaluate potential time-varying effects, we examined whether associations for these significant metabolites varied over follow-up by including interaction terms with follow-up time dichotomized at 15 years. Third, to assess robustness to potential batch effects, risk prediction models were additionally adjusted for a batch indicator variable forced into all models. All analyses were performed using Stata 18 (StataCorp LLC, College Station, TX).

RESULTS

Baseline Characteristics

The mean age of the 3759 participants was 53.5 (SD, 5.7) years, of whom 59.5% were female, and 62.1% were Black participants (Table 1). Compared with those who did not develop PAD during a median follow-up of 27 years (n=3301), those who developed PAD without CLI (n=323) were older, had lower educational attainment, higher cumulative smoking exposure and systolic blood pressure, and a greater prevalence of diabetes. Patterns were generally similar among those who developed CLI (n=135), but unique observations included a much higher prevalence of diabetes (43.7% versus 11.7% among those without PAD), higher systolic blood pressure, and lower estimated glomerular filtration rate. Total cholesterol levels were similar across the 3 groups.

Causes Associated With Incident PAD

A total of 13 metabolites, spanning 4 superpathways and 8 subpathways, were significantly associated with incident PAD ($P<2.04\times 10^{-3}$) after adjusting for traditional risk factors (Figure 1). Among these 13 metabolites,

11 were inversely associated with incident PAD (eg, 1,5-anhydroglucitol [1,5-AG]), whereas 2 were positively associated with incident PAD (eg, mannose; Figure 1). Most (6/13) belonged to the lipid superpathway, predominantly lysolipids such as 2-stearoylglycerophosphocholine (2-stearoyl-glycerophosphocholine) and 1-palmitoleoylglycerophosphocholine (16:1), and the acylcarnitine palmitoylcarnitine (Table 2). The remaining metabolites were 3 carbohydrate metabolites (glucose, 1,5-AG, and mannose), 2 gamma-glutamyl peptides (gamma-glutamylthreonine and gamma-glutamyltyrosine), and 2 cofactors (bilirubin and gamma-carboxyethyl hydroxychroman [CEHC]). Mannose had the strongest positive association (HR per SD, 1.33 [95% CI, 1.16–1.52]; $P=5.51\times 10^{-5}$). The strongest inverse association was for 1,5-AG (HR per SD, 0.78 [95% CI, 0.72–0.85]; $P=4.60\times 10^{-8}$). Pairwise correlations among the metabolites were mostly weak to modest ($|r|<0.4$); however, some metabolites within the same pathway were modestly to strongly correlated with each other (eg, gamma-glutamylthreonine and gamma-glutamyltyrosine $r=0.73$, and 2-stearoyl-glycerophosphocholine and 1-palmitoleoylglycerophosphocholine $r=0.65$; Figure S4).

Causes Associated With Incident CLI

Eighteen metabolites across 5 superpathways and 6 subpathways were significantly associated with incident CLI ($P<2.83\times 10^{-3}$) after adjustment (Figure 2). Again, correlations were modest to strong among some metabolites within the same pathway (Figure S5). The following 7 metabolites overlapped with PAD: 1,5-AG, glucose, gamma-glutamyltyrosine, gamma-glutamylthreonine, 1-pentadecanoyl-glycerophosphocholine (15:0), 2-stearoylglycerophosphocholine, and 1-docosapentaenoylglycerophosphocholine (22:5n3), with similar directions of association observed for both outcomes. Eleven metabolites were associated with CLI only. Among these, homocitrulline showed the strongest positive association (HR per SD, 1.44 [95% CI, 1.19–1.73]), whereas 1-linoleoylglycerol showed the strongest inverse association (HR per SD, 0.70 [95% CI, 0.57–0.85]; Table 3). After accounting for correlated residuals between PAD and CLI, the 11 metabolites identified only for CLI, together with 1,5-AG and gamma-glutamyl peptides, showed significantly stronger associations with CLI than with PAD (Table S1).

Risk Prediction of PAD and CLI

Incorporating the significant metabolites into traditional risk models improved risk prediction for both PAD and CLI. For PAD, the C statistic increased from 0.770 to 0.786 (Δ C statistic, 0.017 [95% CI, 0.008–0.025]), and for CLI, it increased from 0.823 to 0.864 (Δ C statistic, 0.041 [95% CI, 0.018–0.064]; Table 4). The

Table 1. Baseline Characteristics of Visit 1 ARIC Participants With Metabolomic Profiling Included in the Analysis

Characteristics	Total N=3759	No PAD n=3301	PAD without CLI n=323	PAD with CLI n=135
Age, y	53.5 (5.7)	53.3 (5.7)	55.1 (5.9)	54.6 (5.7)
Female	2236 (59.5%)	1999 (60.6%)	163 (50.5%)	74 (54.8%)
Black participants	2333 (62.1%)	2050 (62.1%)	180 (55.7%)	103 (76.3%)
Field center				
Forsyth County	582 (15.5%)	515 (15.6%)	59 (18.3%)	8 (5.9%)
Jackson	2206 (58.7%)	1935 (58.6%)	170 (52.6%)	101 (74.8%)
Minneapolis	510 (13.6%)	456 (13.8%)	42 (13.0%)	12 (8.9%)
Washington County	461 (12.3%)	395 (12.0%)	52 (16.1%)	14 (10.4%)
Education level*				
Basic	1203 (32.0%)	1001 (30.3%)	134 (41.5%)	68 (50.4%)
Intermediate	1281 (34.1%)	1143 (34.6%)	103 (31.9%)	35 (25.9%)
Advanced	1268 (33.7%)	1150 (34.8%)	86 (26.6%)	32 (23.7%)
Smoking (pack-years)	13.1 (19.5)	12.3 (19.3)	20.6 (20.2)	14.5 (19.1)
Physical activity (sport index)	2.3 (0.8)	2.3 (0.8)	2.3 (0.8)	2.2 (0.7)
Systolic BP, mm Hg	124.7 (20.7)	124.0 (20.4)	127.6 (21.3)	135.8 (23.4)
Total cholesterol, mmol/L	5.5 (1.1)	5.5 (1.1)	5.6 (1.2)	5.6 (1.3)
HDL cholesterol, mmol/L	1.4 (0.4)	1.4 (0.5)	1.3 (0.4)	1.2 (0.3)
Diabetes	508 (13.5%)	387 (11.7%)	62 (19.2%)	59 (43.7%)
Prevalent CVD	314 (8.8%)	250 (7.6%)	51 (15.8%)	13 (9.6%)
Antihypertension medication use	1338 (35.6%)	1130 (34.3%)	137 (42.4%)	71 (52.6%)
Lipid-lowering medication use	76 (2.0%)	62 (1.9%)	12 (3.8%)	2 (1.5%)
eGFR, mL/min per 1.73 m ²	100.5 (14.5)	101.0 (14.1)	97.8 (15.7)	96.6 (19.5)
Batch				
1	1768 (47.0%)	1565 (47.4%)	135 (41.8%)	68 (50.4%)
2	1991 (53.0%)	1736 (52.6%)	188 (58.2%)	67 (49.6%)

Continuous variables are expressed as mean (SD) and categorical variables as n (%). Participants were without prevalent PAD. Prevalent CVD: prevalent coronary heart disease and stroke. eGFR was calculated using Chronic Kidney Disease Epidemiology Collaboration 2021 equation. ARIC indicates Atherosclerosis Risk in Communities; BP, blood pressure; CKD, chronic kidney disease; CLI, critical limb ischemia; CVD, cardiovascular disease; eGFR, estimated glomerular filtration rate; HDL, high-density lipoprotein; and PAD, peripheral artery disease.

*Education level: basic=less than completed high school; intermediate=completed high school or equivalent; advanced—at least some college. Missing values in the analytic sample were as follows: educational level, n=7; smoking, n=88; diabetes, n=7; total cholesterol, n=46; HDL cholesterol, n=44; physical activity, n=16; antihypertensive medication use, n=3; and lipid-lowering medication use, n=38. Other covariates shown in Table 1 had no missing values.

categorical net reclassification improvement was also significantly positive for both PAD and CLI (0.053 [95% CI, 0.018–0.083] and 0.176 [95% CI, 0.085–0.268], respectively). Similarly, the continuous net reclassification improvement was significantly positive for both PAD and CLI.

Sensitivity Analysis

Our results were generally consistent across sensitivity analyses. When follow-up was restricted to 15 years, almost all metabolites identified as significant in the primary analyses for PAD and CLI remained significant, with consistent directions and similar magnitudes of association (Tables S2 and S3). In sensitivity analyses stratifying follow-up at 15 years, although their magnitude varied

for some metabolites, the directionality of their associations was consistent during earlier and later follow-up (Tables S4 and S5). Finally, additional adjustment for batch indicators in risk prediction models did not change estimates of discrimination or reclassification for either PAD or CLI (Table S6).

DISCUSSION

In this large, community-based cohort followed for >2 decades, we identified 13 circulating metabolites associated with incident PAD, with signals concentrated in 4 superpathways: lipid, carbohydrate, peptide, and cofactors/vitamins. Eleven metabolites, including several lysolipids, were inversely related to PAD risk. We also found 18 metabolites associated with CLI, of which 11 were

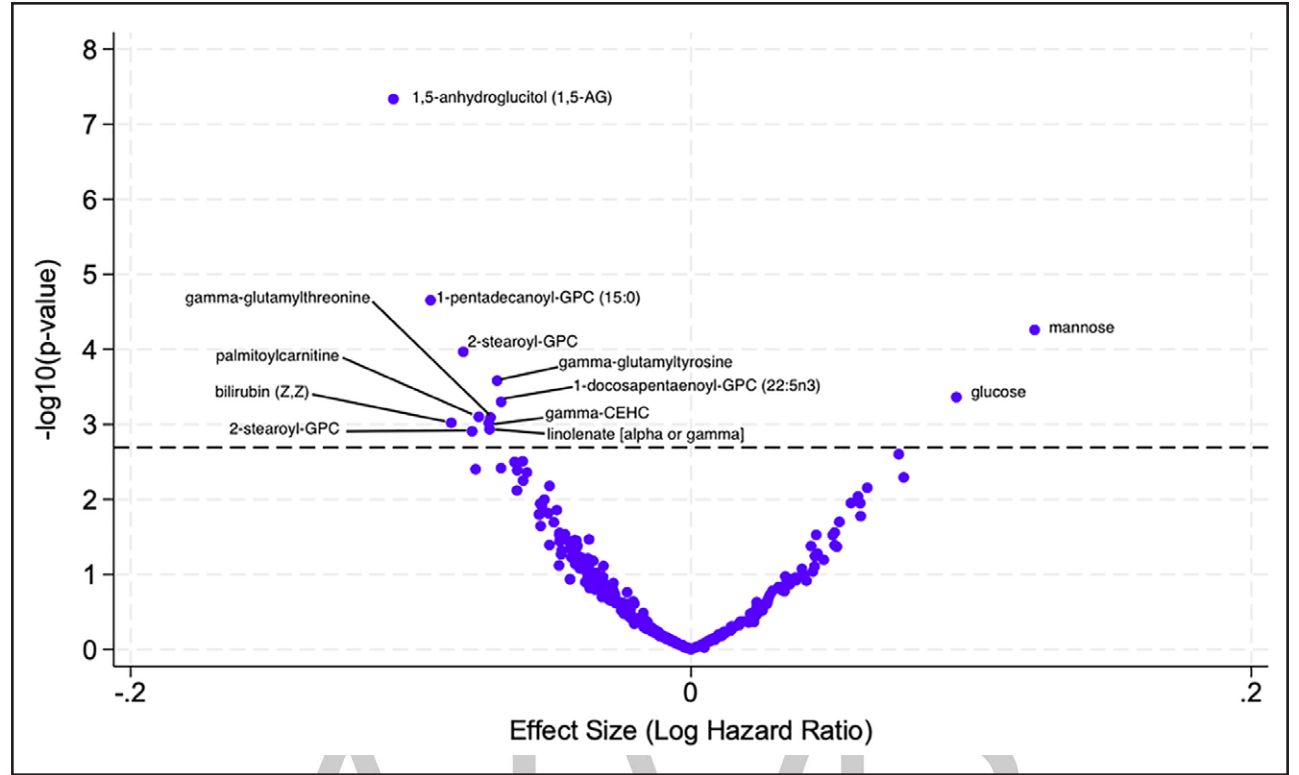


Figure 1. Volcano plot of associations for individual metabolites and incident peripheral artery disease (PAD). Horizontal dashed line represents the Benjamin-Hochberg false discovery rate–adjusted threshold of 2.04×10^{-3} . CEHC indicates carboxyethyl hydroxychroman; and GPC, glycerophosphocholine.

associated specifically with CLI in the primary analyses (eg, homocitrulline, arabinonate, and 1-linoleoylglycerol). When PAD and CLI were modeled jointly while accounting for correlated outcomes, several metabolites, including 1,5-AG and gamma-glutamyl peptides, showed significantly stronger associations with CLI than with

Table 2. Metabolites and Pathways Associated With Incident PAD

Metabolite	Superpathway	Subpathway	HR (95% CI)	P value
1,5-AG	Carbohydrate	Glycolysis, gluconeogenesis, and pyruvate	0.78 (0.72–0.85)	4.60×10^{-8}
1-Pentadecanoyl-GPC (15:0)	Lipid	Lysolipid	0.81 (0.73–0.89)	2.22×10^{-5}
Mannose	Carbohydrate	Fructose, mannose, and galactose	1.33 (1.16–1.52)	5.51×10^{-5}
2-Stearoyl-GPC	Lipid	Lysolipid	0.83 (0.75–0.91)	1.08×10^{-4}
Gamma-glutamyltyrosine	Peptide	Gamma-glutamyl amino acid	0.85 (0.78–0.93)	2.61×10^{-4}
Glucose	Carbohydrate	Glycolysis, gluconeogenesis, and pyruvate	1.24 (1.10–1.40)	4.35×10^{-4}
1-Docosapentaenoyl-GPC (22:5n3)	Lipid	Lysolipid	0.86 (0.78–0.93)	5.01×10^{-4}
Palmitoylcarnitine	Lipid	Fatty-acid metabolism (acylcarnitine)	0.84 (0.76–0.93)	7.95×10^{-4}
Gamma-glutamylthreonine	Peptide	Gamma-glutamyl amino acid	0.85 (0.77–0.93)	8.10×10^{-4}
Gamma-CEHC	Cofactors and Vitamins	Tocopherol metabolism	0.85 (0.77–0.93)	9.53×10^{-4}
Bilirubin (Z,Z)	Cofactors and vitamins	Hemoglobin and porphyrin	0.82 (0.73–0.92)	9.54×10^{-4}
Linolenate (alpha or gamma)	Lipid	Polyunsaturated fatty acid (n3 and n6)	0.85 (0.77–0.94)	1.16×10^{-3}
1-Palmitoleoyl-GPC (16:1)	Lipid	Lysolipid	0.84 (0.75–0.93)	1.23×10^{-3}

Associations of individual metabolites with incident PAD were assessed using Cox proportional hazards models. Multiple testing was addressed using the Benjamini-Hochberg FDR procedure, with an FDR-adjusted significance threshold of $P < 2.04 \times 10^{-3}$. Participants with prevalent PAD were excluded at baseline. HRs represent associations per 1-SD increase in metabolite levels. HR < 1.00 indicates an inverse association (higher metabolite levels associated with lower risk), whereas HR > 1.00 indicates a positive association (higher metabolite levels associated with higher risk). For example, an HR of 0.85 for gamma-CEHC indicates that a 1-SD higher level of gamma-CEHC was associated with a 15% lower risk of PAD. Models were adjusted for age, sex, race-center, batch, education level, smoking, physical activity, diabetes, systolic blood pressure, total cholesterol, high-density lipoprotein cholesterol, prevalent cardiovascular disease, medication use (antihypertensive and lipid-lowering), and estimated glomerular filtration rate. Metabolites are arranged in order of statistical significance. 1,5-AG indicates 1,5-anhydroglucitol; CEHC, carboxyethyl hydroxychroman; FDR, false discovery rate; GPC, glycerophosphocholine; HR, hazard ratio; and PAD, peripheral artery disease.

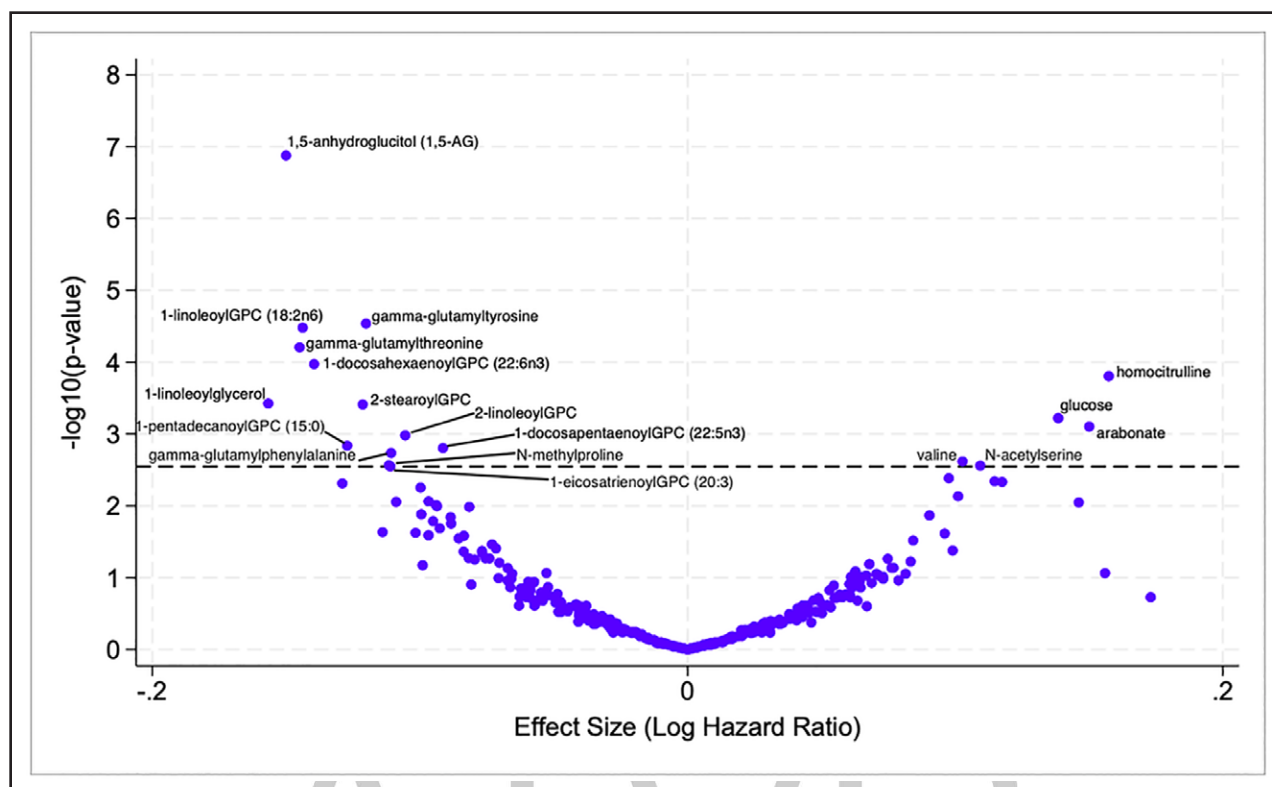


Figure 2. Volcano plot of associations for individual metabolites and incident critical limb ischemia (CLI).

The horizontal dashed line represents the Benjamini-Hochberg false discovery rate-adjusted threshold of 2.83×10^{-3} . GPC indicates glycerophosphocholine.

PAD. Addition of these metabolite panels improved risk discrimination for both outcomes (ΔC statistics ≈ 0.02 for PAD and ≈ 0.04 for CLI) beyond traditional atherosclerotic risk factors.

Our study advances the limited literature on metabolomics and PAD by identifying distinct metabolic signatures associated with incident PAD and its severe manifestation, CLI, decades before clinical onset. We observed both shared and CLI-predominant associations, underscoring the importance of considering CLI as a distinct outcome rather than simply advanced PAD. Several metabolites identified in our analyses, including mannose, gamma-glutamyl peptides, gamma-CEHC, 1-linoleoylglycerol, homocitrulline, and arabinonate, have not been previously linked to PAD or CLI in prospective epidemiological studies, despite prior evidence implicating some of these metabolites in related cardiometabolic or experimental contexts.

Although the metabolites identified for PAD span 4 metabolomic superpathways, their biological interpretation points to 3 biological processes potentially involved in the development of PAD: glycemic control, redox balance, and lipid remodeling (dynamic changes in cell membranes that help maintain homeostasis⁴⁸). Importantly, several metabolites within the same pathway were modestly to strongly correlated with each other. Nonetheless, these findings reflect population-level perturbations and

may not necessarily represent coordinated metabolite changes within individuals.

It is not surprising that several metabolites involved in glucose metabolism emerged, given that diabetes is a leading risk factor for PAD. For example, 1,5-AG, a well-established marker that is inversely related to short-term hyperglycemia,⁴⁹ was among the metabolites most strongly associated with PAD in our study. We also identified mannose, another sugar-related metabolite, as strongly associated with PAD. Beyond its traditional link to glucose metabolism, elevated mannose has been shown to potentially weaken the fibrous cap of atheroma and increase plaque vulnerability, even among individuals with normal glycemic indices, by promoting abnormal *N*-glycosylation and protein misfolding.^{50,51}

We observed inverse associations of several metabolites involved in antioxidant defense, including gamma-glutamyl peptides, bilirubin, and a vitamin E metabolite (gamma-CEHC) with PAD.⁵² Lower levels of gamma-glutamyl peptides reflect impaired detoxification of reactive oxygen species,⁵³ and lower bilirubin and vitamin E derivatives indicate reduced protection against membrane lipid peroxidation.^{54,55} To our knowledge, neither gamma-glutamyl peptides nor gamma-CEHC has been previously linked to cardiovascular disease outcomes. Our results suggest that impaired redox homeostasis is particularly important for the

Table 3. Metabolites and Pathways Associated With Incident CLI

Metabolite	Superpathway	Subpathway	HR (95% CI)	P value
1,5-AG	Carbohydrate	Glycolysis, gluconeogenesis, and pyruvate	0.71 (0.62–0.80)	1.33×10 ⁻⁷
Gamma-glutamyltyrosine	Peptide	Gamma-glutamyl amino acid	0.76 (0.67–0.86)	2.90×10 ⁻⁵
1-Linoleoyl-GPC (18:2n6)	Lipid	Lysolipid	0.72 (0.61–0.84)	3.31×10 ⁻⁵
Gamma-glutamylthreonine	Peptide	Gamma-glutamyl amino acid	0.72 (0.61–0.84)	6.19×10 ⁻⁵
1-Docosahexaenoyl-GPC (22:6n3)	Lipid	Lysolipid	0.73 (0.62–0.85)	1.06×10 ⁻⁴
Homocitrulline	Amino acid	Urea cycle; arginine and proline Met.	1.44 (1.19–1.73)	1.56×10 ⁻⁴
1-Linoleoylglycerol	Lipid	Monoacylglycerol	0.70 (0.57–0.85)	3.76×10 ⁻⁴
2-Stearoyl-GPC	Lipid	Lysolipid	0.76 (0.65–0.88)	3.88×10 ⁻⁴
Glucose	Carbohydrate	Glycolysis, gluconeogenesis, and pyruvate	1.38 (1.15–1.65)	6.02×10 ⁻⁴
Arabonate	Cofactors and vitamins	Ascorbate and aldarate Met.	1.41 (1.15–1.73)	7.87×10 ⁻⁴
2-Linoleoyl-GPC	Lipid	Lysolipid	0.78 (0.68–0.91)	1.04×10 ⁻³
1-Pentadecanoyl-GPC (15:0)	Lipid	Lysolipid	0.75 (0.62–0.89)	1.45×10 ⁻³
1-Docosapentaenoyl-GPC (22:5n3)	Lipid	Lysolipid	0.81 (0.71–0.92)	1.58×10 ⁻³
Gamma-glutamylphenylalanine	Peptide	Gamma-glutamyl amino acid	0.77 (0.66–0.91)	1.83×10 ⁻³
Valine	Amino acid	Leucine, isoleucine, and valine Met.	1.27 (1.09–1.48)	2.41×10 ⁻³
N-methylproline	Amino acid	Urea cycle; arginine and proline Met.	0.77 (0.65–0.91)	2.72×10 ⁻³
N-acetylserine	Amino acid	Glycine, serine, and threonine Met.	1.29 (1.09–1.52)	2.76×10 ⁻³
1-Eicosatrienoyl-GPC (20:3)	Lipid	Lysolipid	0.77 (0.65–0.92)	2.81×10 ⁻³

Associations of individual metabolites with incident CLI were assessed using Cox proportional hazards models. Multiple testing was addressed using the Benjamini-Hochberg FDR procedure, with an FDR-adjusted significance threshold of $P < 2.83 \times 10^{-3}$. Participants with prevalent PAD were excluded at baseline. HRs represent associations per 1-SD increase in metabolite levels. HR <1.00 indicates an inverse association (higher metabolite levels associated with lower risk), whereas HR >1.00 indicates a positive association (higher metabolite levels associated with higher risk). For example, an HR of 0.71 for 1,5-AG indicates that a 1-SD higher level of 1,5-AG was associated with a 29% lower risk of PAD. Models were adjusted for age, sex, race-center, batch, education level, smoking, physical activity, diabetes, systolic blood pressure, total cholesterol, high-density lipoprotein cholesterol, prevalent cardiovascular disease, medication use (antihypertensive and lipid-lowering), and estimated glomerular filtration rate. Metabolites are arranged in order of statistical significance. 1,5-AG indicates 1,5-anhydroglucitol; CLI, critical limb ischemia; FDR, false discovery rate; GPC, glycerophosphocholine; HR, hazard ratio; Met, metabolism; and PAD, peripheral artery disease.

development of PAD. This conclusion is supported by a randomized controlled trial in older adults with PAD that demonstrated that cocoa supplementation rich in flavanols with antioxidant properties improved walking performance and skeletal muscle mitochondrial function.⁵⁶

The association of metabolites linked to lipid remodeling with PAD in our study aligns with prior studies implicating lysophospholipids and acylcarnitines, amphipathic

byproducts of incomplete lipid metabolism, in atherosclerotic cardiovascular disease, including PAD.^{23,57} Lysophospholipids, generated from the hydrolysis of phosphatidylcholines in ApoB-containing lipoproteins,⁵⁸ promote vascular inflammation and oxidative stress that contribute to atherosclerotic plaques and PAD.⁵⁹ Acylcarnitines reflect mitochondrial fatty-acid oxidation and accumulate in established PAD.^{23,58} In our study, however, lower palmitoylcarnitine levels were associated with

Table 4. Evaluation of Risk Prediction Models

Performance metric	PAD		CLI	
	Traditional	+13 metabolites	Traditional	+18 metabolites
C-index (95% CI)	0.770 (0.751–0.789)	0.786 (0.769–0.804)	0.823 (0.793–0.852)	0.864 (0.832–0.896)
C-index (95% CI)	...	0.017 (0.008–0.025)	...	0.041 (0.018–0.064)
[ΔC] P value	...	<0.001	...	<0.001
Categorical NRI (95% CI)	...	0.053 (0.018–0.083)	...	0.176 (0.085–0.268)
Continuous NRI (95% CI)	...	0.317 (0.208–0.426)	...	0.618 (0.446–0.790)

Traditional risk factor model included age, sex, smoking, systolic blood pressure, diabetes, total cholesterol, and HDL cholesterol. PAD-related metabolites=gamma-glutamyltyrosine, gamma-glutamylthreonine, 1,5-anhydroglucitol, glucose, mannose, linolenate (alpha or gamma), palmitoylcarnitine, 1-pentadecanoylglycerophosphocholine (15:0), 2-stearoylglycerophosphocholine, 1-docosapentaenoylglycerophosphocholine (22:5n3), gamma-CEHC, bilirubin (Z,Z). CLI-related metabolites=1,5-anhydroglucitol, gamma-glutamylphenylalanine, gamma-glutamyltyrosine, gamma-glutamylthreonine, 1-docosahexaenoylglycerophosphocholine (22:6n3), 1-linoleoylglycerophosphocholine (18:2n6), 2-linoleoylglycerophosphocholine, 1-linoleoylglycerol (1-monolinolein), 2-stearoylglycerophosphocholine, arabonate, glucose, 1-docosapentaenoylglycerophosphocholine (22:5n3), homocitrulline, 1-pentadecanoylglycerophosphocholine (15:0), 1-eicosatrienoylglycerophosphocholine (20:3), N-acetylserine, valine, N-methylproline. NRI risk categories: PAD: <8%, 8% to 23%, >23%; CLI: <2%, 2% to 6%, >6%. CEHC indicates carboxyethyl hydroxychroman; CLI, critical limb ischemia; HDL, high-density lipoprotein; NRI, net reclassification index; and PAD, peripheral artery disease.

a higher risk of PAD, suggesting that the associations involving acylcarnitines may not be monotonic and may vary by disease stage. Similar stage-dependent patterns have been observed for related lipid metabolites such as sphingolipids and ceramides, which exhibit a biphasic trajectory in the development of diabetes, with reduced levels in prediabetes and increased levels in overt diabetes.⁶⁰ Taken together, these findings support a role for chronic low-grade inflammation in PAD, as oxidative stress, impaired antioxidant defenses, and lipid-mediated vascular injury are closely linked to inflammatory activation and endothelial dysfunction.

Despite fewer cases and lower statistical power, we found a greater number of metabolites associated with CLI, compared with overall PAD. For example, homocitrulline, amino acid-derived metabolites, arabonate, 1-linoleoylglycerol, and polyunsaturated lysophospholipids were preferentially associated with CLI. Although CLI is a severe form of PAD, clinical observations and data indicate that the development of CLI is not merely a gradual progression from mild to severe PAD. Indeed, CLI is the first manifestation of PAD in some patients.^{61,62} In addition, the development of CLI likely represents complex pathophysiology, including ischemia, inflammation, infection, wound healing, and collateral creation. For example, alterations in monoacylglycerol species, such as 1-linoleoylglycerol and polyunsaturated lysophospholipids like 1-docosahexaenoyl-glycerophosphocholine (22:6n3), imply ongoing breakdown of cell membranes and compromised mitochondrial lipid oxidation.⁶³ In addition, arabonate, although not known to play a direct role in human pathology, may reflect broader oxidative or microbial carbohydrate metabolism.⁶⁴

Our findings highlight the circulating metabolites as early indicators of vascular health. Indeed, several metabolites in the present study were prognostic of PAD and CLI >10 to 20 years. The implicated pathways of dysregulated glycemic control, impaired redox balance, and altered lipid remodeling also point to modifiable targets for prevention. For example, metabolites such as mannose reinforce the importance of optimizing glycemic control even ahead of overt diabetes, whereas inverse associations with gamma-glutamyl peptides, bilirubin, and gamma-CEHC suggest that enhancing antioxidant and mitochondrial function may represent targets for preventive strategies.⁵⁶ Notably, gamma-CEHC is a metabolite of gamma-tocopherol, whereas alpha-tocopherol, the predominant form in most standard vitamin E supplements, has been shown to suppress circulating gamma-tocopherol.⁶⁵ Our findings may have important implications for interpreting largely negative clinical trials of alpha-tocopherol supplementation for vascular health.^{66–68} The identification of unique metabolites for CLI supports a pathophysiology that extends beyond mere progression of PAD and underscores the need for improved identification of individuals at risk for

limb-threatening ischemia. This finding is relevant given the documented upward trend in CLI-related hospitalizations over the past decade, particularly among younger adults.⁶¹

Several limitations of our study warrant consideration. First, metabolite levels were measured at a single time point, and thus their temporal stability or changes over time could not be assessed. Nevertheless, it is remarkable that other prospective studies and ours have shown that baseline single-time metabolomic profiles can still predict long-term disease risk.^{69–74} These observations suggest that a single-time metabolic profile may capture underlying metabolic milieu predisposing to diseases, including PAD and CLI. Second, serum samples from ARIC visit 1 were stored at -80°C for >2 decades. Because samples were collected before PAD or CLI events, storage-related changes would be expected to introduce nondifferential misclassification in metabolite concentrations. Such measurement error, whether upward or downward, typically attenuates observed associations relative to the true underlying associations, if any. Therefore, we may have missed associations for some storage-sensitive metabolites. Third, the analyzed cohort included only Black and White participants, which may limit generalizability to other racial groups. Fourth, PAD and CLI were ascertained using hospital *International Classification of Diseases* codes, which may be subject to misclassification. In addition, milder cases not requiring hospitalization were not captured. Nonetheless, such a misclassification usually results in conservative estimates. Fifth, the improvement in risk prediction with the addition of metabolite panels should be externally validated. Finally, although we rigorously adjusted for a wide range of covariates, there remains the possibility of residual confounding. Despite these limitations, our study also offers several notable strengths, including the use of a large, well-characterized, community-based cohort; prospective, untargeted metabolomic profiling with long-term follow-up over a few decades; and rigorous statistical analyses, including prediction statistics.

We linked circulating metabolites measured decades before clinical onset to future development of PAD and, importantly, CLI. These metabolic signatures implicate pathways related to glycemic control, redox balance, and lipid remodeling, highlighting metabolic processes that precede overt disease. The identification of CLI-unique metabolites supports growing evidence that CLI represents a biologically distinct, high-risk phenotype. Together, these findings have implications for early identification and prevention of PAD and its severe form, CLI.

ARTICLE INFORMATION

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Disclosures

None.

Supplemental Material

Tables S1–S6

Figures S1–S5

Major Resources Table

ARRIVE Guidelines

REFERENCES

- Wilkins LR, Sabri SS, Misra S. The 2024 ACC/AHA/AACVPR/APMA/ABC/SCAI/SVM/SVN/SVS/SIR/VES guideline for the management of lower extremity peripheral artery disease: pertinent points for the interventional radiologist. *J Vasc Interv Radiol*. 2024;35:1743–1751. doi: 10.1016/j.jvir.2024.08.024
- Shu J, Santulli G. Update on peripheral artery disease: epidemiology and evidence-based facts. *Atherosclerosis*. 2018;275:379–381. doi: 10.1016/j.atherosclerosis.2018.05.033
- Song P, Rudan D, Zhu Y, Fowkes FJ, Rahimi K, Fowkes FGR, Rudan I. Global, regional, and national prevalence and risk factors for peripheral artery disease in 2015: an updated systematic review and analysis. *Lancet Glob Health*. 2019;7:e1020–e1030. doi: 10.1016/S2214-109X(19)30255-4
- McDermott MM. Functional impairment in peripheral artery disease and how to improve it in 2013. *Curr Cardiol Rep*. 2013;15:1–8. doi: 10.1007/s11886-013-0347-5
- Beckman JA, Duncan MS, Damrauer SM, Wells QS, Barnett JV, Wasserman DH, Bedimo RJ, Butt AA, Marconi VC, Sico JJ, et al. Microvascular disease, peripheral artery disease, and amputation. *Circulation*. 2019;140:449–458. doi: 10.1161/CIRCULATIONAHA.119.040672
- Fowkes FGR, Murray GD, Butcher I, Heald CL, Lee RJ, Chambless LE, Folsom AR, Hirsch AT, Dramaix M, deBacker G, et al; Ankle Brachial Index Collaboration. Ankle brachial index combined with Framingham Risk Score to predict cardiovascular events and mortality: a meta-analysis. *JAMA*. 2008;300:197–208. doi: 10.1001/jama.300.2.197
- Gerhard-Herman MD, Gornik HL, Barrett C, Barshes NR, Corriere MA, Drachman DE, Fleisher LA, Fowkes FGR, Hamburg NM, Kinlay S, et al. 2016 AHA/ACC guideline on the management of patients with lower extremity peripheral artery disease: a report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. *J Am Coll Cardiol*. 2017;69:e71–e126. doi: 10.1016/j.jacc.2016.11.007
- Kengne AP, Echouffo-Tcheugui JB. Differential burden of peripheral artery disease. *Lancet Glob Health*. 2019;7:e980–e981. doi: 10.1016/S2214-109X(19)30293-1
- Go AS, Mozaffarian D, Roger VL, Benjamin EJ, Berry JD, Blaha MJ, Dai S, Ford ES, Fox CS, Franco S, et al; American Heart Association Statistics Committee and Stroke Statistics Subcommittee. Heart disease and stroke statistics—2014 update: a report from the American Heart Association. *Circulation*. 2013;129:e28–e292. doi: 10.1161/01.cir.0000441139.02102.80
- Ding N, Sang Y, Chen J, Ballew SH, Kalbaugh CA, Salameh MJ, Blaha MJ, Allison M, Heiss G, Selvin E, et al. Cigarette smoking, smoking cessation, and long-term risk of 3 major atherosclerotic diseases. *J Am Coll Cardiol*. 2019;74:498–507. doi: 10.1016/j.jacc.2019.05.049
- Giannopoulos S, Armstrong EJ. Diabetes mellitus: an important risk factor for peripheral vascular disease. *Expert Rev Cardiovasc Ther*. 2020;18:131–137. doi: 10.1080/14779072.2020.1736562
- Tzoulaki I, Murray GD, Lee AJ, Rumley A, Lowe GD, Fowkes FGR. C-reactive protein, interleukin-6, and soluble adhesion molecules as predictors of progressive peripheral atherosclerosis in the general population: Edinburgh Artery Study. *Circulation*. 2005;112:976–983. doi: 10.1161/CIRCULATIONAHA.104.513085
- Aboyans V, Criqui MH, Denenberg JO, Knoke JD, Ridker PM, Fronek A. Risk factors for progression of peripheral arterial disease in large and small vessels. *Circulation*. 2006;113:2623–2629. doi: 10.1161/CIRCULATIONAHA.105.608679
- Aboyans V, McClelland RL, Allison MA, McDermott MM, Blumenthal RS, Macura K, Criqui MH. Lower extremity peripheral artery disease in the absence of traditional risk factors. The Multi-Ethnic Study of Atherosclerosis. *Atherosclerosis*. 2011;214:169–173. doi: 10.1016/j.atherosclerosis.2010.10.011
- Tarvainen S, Wirth G, Juusola G, Hautero O, Kalliokoski K, Sjöros T, Nikulainen V, Taavitsainen J, Hytönen J, Frimodig C, et al. Critical limb-threatening ischaemia and microvascular transformation: clinical implications. *Eur Heart J*. 2024;45:255–264. doi: 10.1093/eurheartj/ehad562
- Signorelli SS, Scuto S, Marino E, Xourafa A, Gaudio A. Oxidative stress in peripheral arterial disease (PAD) mechanism and biomarkers. *Antioxidants*. 2019;8:367. doi: 10.3390/antiox8090367
- Nogal A, Alkis T, Lee Y, Kifer D, Hu J, Murphy RA, Huang Z, Wang-Sattler R, Kastenmüller G, Linkohr B, et al. Predictive metabolites for incident myocardial infarction: a two-step meta-analysis of individual patient data from six cohorts comprising 7897 individuals from the Consortium of METabolomics Studies. *Cardiovasc Res*. 2023;119:2743–2754. doi: 10.1093/cvr/cvad147
- Ke C, Pan CW, Zhang Y, Zhu X, Zhang Y. Metabolomics facilitates the discovery of metabolic biomarkers and pathways for ischemic stroke: a systematic review. *Metabolomics*. 2019;15:1–21. doi: 10.1007/s11306-019-1615-1
- Tiedt S, Brandmaier S, Kollmeier H, Duering M, Artati A, Adamski J, Klein M, Liebig T, Holdt LM, Teupser D, et al. Circulating metabolites differentiate acute ischemic stroke from stroke mimics. *Ann Neurol*. 2020;88:736–746. doi: 10.1002/ana.25859
- Azab SM, Zamzam A, Syed MH, Abdin R, Qadura M, Britz-McKibbin P. Serum metabolic signatures of chronic limb-threatening ischemia in patients with peripheral artery disease. *J Clin Med*. 2020;9:1877. doi: 10.3390/jcm9061877
- Hernández-Aguilera A, Fernández-Arroyo S, Cabre N, Luciano-Mateo F, Baiges-Gaya G, Fibla M, Martín-Paredero V, Menéndez JA, Camps J, Joven J. Plasma energy-balance metabolites discriminate asymptomatic patients with peripheral artery disease. *Mediators Inflamm*. 2018;2018:2760272. doi: 10.1155/2018/2760272
- Huang CC, McDermott MM, Liu K, Kuo CH, Wang SY, Tao H, Tseng YJ. Plasma metabolomic profiles predict near-term death among individuals with lower extremity peripheral arterial disease. *J Vasc Surg*. 2013;58:989–996. doi: 10.1016/j.jvs.2013.04.022
- Ismaeel A, Franco ME, Lavado R, Papoutsis E, Casale GP, Fuglestad M, Mietus CJ, Haynatzki GR, Smith RS, Bohannon WT, et al. Altered metabolomic profile in patients with peripheral artery disease. *J Clin Med*. 2019;8:1463. doi: 10.3390/jcm8091463
- Lee J, Gilliland T, Koyama S, Nakao T, Dron J, Lannery K, Wong M, Peloso GM, Hornsby W, Natarajan P. Integrative metabolomics differentiate coronary artery disease, peripheral artery disease, and venous thromboembolism risks. *Arterioscler Thromb Vasc Biol*. 2024;44:2108–2117. doi: 10.1161/ATVBAHA.124.321282
- Razquin C, Ruiz-Canela M, Toledo E, Clish CB, Guasch-Ferré M, García-Gavián JF, Wittenbecher C, Alonso-Gómez A, Fitó M, Liang L, et al. Circulating amino acids and risk of peripheral artery disease in the PRE-DIMED trial. *Int J Mol Sci*. 2022;24:270. doi: 10.3390/ijms24010270
- Zagura M, Kals J, Kilk K, Serg M, Kampus P, Eha J, Soomets U, Zilmer M. Metabolomic signature of arterial stiffness in male patients with peripheral arterial disease. *Hypertens Res*. 2015;38:840–846. doi: 10.1038/hr.2015.71
- Wright JD, Folsom AR, Coresh J, Sharrett AR, Couper D, Wagenknecht LE, Mosley TH, Ballantyne CM, Boerwinkle EA, Rosamond WD, et al. The ARIC (Atherosclerosis Risk in Communities) study: JACC Focus Seminar 3/8. *J Am Coll Cardiol*. 2021;77:2939–2959. doi: 10.1016/j.jacc.2021.04.035
- Bernard L, Zhou L, Surapaneni A, Chen J, Rebholz CM, Coresh J, Yu B, Boerwinkle E, Schlosser P, Grams ME. Serum metabolites and kidney

- outcomes: the Atherosclerosis Risk in Communities study. *Kidney Med*. 2022;4:100522. doi: 10.1016/j.xkme.2022.100522
29. Yu B, Heiss G, Alexander D, Grams ME, Boerwinkle E. Associations between the serum metabolome and all-cause mortality among African Americans in the Atherosclerosis Risk in Communities (ARIC) study. *Am J Epidemiol*. 2016;183:650–656. doi: 10.1093/aje/kwv213
 30. Evans AM, DeHaven CD, Barrett T, Mitchell M, Milgram E. Integrated, non-targeted ultrahigh performance liquid chromatography/electrospray ionization tandem mass spectrometry platform for the identification and relative quantification of the small-molecule complement of biological systems. *Anal Chem*. 2009;81:6656–6667. doi: 10.1021/ac901536h
 31. Ohta T, Masutomi N, Tsutsui N, Sakairi T, Mitchell M, Milburn MV, Ryals JA, Beebe KD, Guo L. Untargeted metabolomic profiling as an evaluative tool of fenofibrate-induced toxicology in Fischer 344 male rats. *Toxicol Pathol*. 2009;37:521–535. doi: 10.1177/0192623309336152
 32. Zheng Y, Yu B, Alexander D, Manolio TA, Aguilar D, Coresh J, Heiss G, Boerwinkle E, Nettleton JA. Associations between metabolomic compounds and incident heart failure among African Americans: the ARIC study. *Am J Epidemiol*. 2013;178:534–542. doi: 10.1093/aje/kwt004
 33. Zheng Y, Yu B, Alexander D, Mosley TH, Heiss G, Nettleton JA, Boerwinkle E. Metabolomics and incident hypertension among Blacks: the atherosclerosis risk in communities study. *Hypertension*. 2013;62:398–403. doi: 10.1161/HYPERTENSIONAHA.113.01166
 34. Yu B, Zheng Y, Nettleton JA, Alexander D, Coresh J, Boerwinkle E. Serum metabolomic profiling and incident CKD among African Americans. *Clin J Am Soc Nephrol*. 2014;9:1410–1417. doi: 10.2215/CJN.11971113
 35. Wattanakit K, Folsom AR, Selvin E, Coresh J, Hirsch AT, Weatherley BD. Kidney function and risk of peripheral arterial disease: results from the Atherosclerosis Risk in Communities (ARIC) study. *J Am Soc Nephrol*. 2007;18:629–636. doi: 10.1681/ASN.2005111204
 36. Bekwelem W, Bengtson LG, Oldenburg NC, Winden TJ, Keo HH, Hirsch AT, Duval S. Development of administrative data algorithms to identify patients with critical limb ischemia. *Vasc Med*. 2014;19:483–490. doi: 10.1177/1358863X14559589
 37. Richardson MT, Ainsworth BE, Wu HC, Jacobs DR Jr, Leon AS. Ability of the Atherosclerosis Risk in Communities (ARIC)/Baecke Questionnaire to assess leisure-time physical activity. *Int J Epidemiol*. 1995;24:685–693. doi: 10.1093/ije/24.4.685
 38. Atherosclerosis Risk in Communities Study Exam 1: Derived Variable Dictionary, Version 13. Accessed September 12. https://aric.csc.unc.edu/aric9/sites/default/files/public/visitdocuments/v1/Derived%20Variable%20Dictionary%20Visit%201_0.pdf
 39. Atherosclerosis Risk in Communities Data Book Exam 1: Stroke Data. Accessed September 12. <https://aric.csc.unc.edu/aric9/sites/default/files/public/visitdocuments/v1/STROKE01.pdf>
 40. Atherosclerosis Risk in Communities Study Protocol Manual 11: Sitting Blood Pressure, Version 1.0. https://aric.csc.unc.edu/aric9/sites/default/files/public/visitdocuments/v1/Sitting_Blood_Pressure_and_Postural_Changes_in_Blood_Pressure_and_Heart_Rate_1_11.pdf
 41. Miller WG, Kaufman HW, Levey AS, Straseski JA, Wilhelms KW, Yu HY, Klutts JS, Hilborne LH, Horowitz GL, Lieske J. National Kidney Foundation Laboratory Engagement Working Group recommendations for implementing the CKD-EPI 2021 race-free equations for estimated glomerular filtration rate: practical guidance for clinical laboratories. *Clin Chem*. 2022;68:511–520. doi: 10.1093/clinchem/hvab278
 42. Atherosclerosis Risk in Communities Study Protocol, Manual 8: Lipid And Lipoprotein Determinations, Version 1.0. Accessed September 12. https://aric.csc.unc.edu/aric9/sites/default/files/public/visitdocuments/v1/Lipid_and_Lipoprotein_Determinations.1_8.pdf
 43. Grambsch PM, Therneau TM. Proportional hazards tests and diagnostics based on weighted residuals. *Biometrika*. 1994;81:515–526. doi: 10.1093/biomet/81.3.515
 44. Harrell FE, Califf RM, Pryor DB, Lee KL, Rosati RA. Evaluating the yield of medical tests. *JAMA*. 1982;247:2543–2546.
 45. Harrell FE Jr, Lee KL, Mark DB. Multivariable prognostic models: issues in developing models, evaluating assumptions and adequacy, and measuring and reducing errors. *Stat Med*. 1996;15:361–387. doi: 10.1002/(SICI)1097-0258(19960229)15:4<361::AID-SIM168>3.0.CO;2-4
 46. Pencina MJ, D'Agostino Sr RB, Steyerberg EW. Extensions of net reclassification improvement calculations to measure usefulness of new biomarkers. *Stat Med*. 2011;30:11–21. doi: 10.1002/sim.4085
 47. D'Agostino Sr RB, Pencina MJ, Massaro JM, Coady S. Cardiovascular disease risk assessment: insights from Framingham. *Glob Heart*. 2013;8:11–23. doi: 10.1016/j.gheart.2013.01.001
 48. Cho YK, Lee S, Lee J, Doh J, Park JH, Jung YS, Lee YH. Lipid remodeling of adipose tissue in metabolic health and disease. *Exp Mol Med*. 2023;55:1955–1973. doi: 10.1038/s12276-023-01071-4
 49. Selvin E, Rawlings A, Lutsey P, Maruthur N, Pankow JS, Steffes M, Coresh J. Association of 1,5-anhydroglucitol with cardiovascular disease and mortality. *Diabetes*. 2016;65:201–208. doi: 10.2337/db15-0607
 50. Ferrannini E, Marx N, Andreini D, Campi B, Saba A, Gorini M, Ferrannini G, Milzi A, Magnoni M, Maseri A, et al; CAPIRE Investigators. Mannose as a biomarker of coronary artery disease: angiographic evidence and clinical significance. *Int J Cardiol*. 2022;346:86–92. doi: 10.1016/j.ijcard.2021.11.038
 51. Fortin E, Campi B, Ferrannini E, Mari A, Mellbin LG, Norhammar A, Näsman P, Rydén L, Saba A, Ferrannini G. High mannose correlates with surrogate indexes of insulin resistance and is associated with an increased risk of cardiovascular events independently of glycemic status and traditional risk factors. *Diabetes Care*. 2024;47:246–251. doi: 10.2337/dc23-0870
 52. Gardner AW, Parker DE, Montgomery PS, Sosnowska D, Casanegra AI, Ungvari Z, Csiszar A, Sonntag WE. Greater endothelial apoptosis and oxidative stress in patients with peripheral artery disease. *Int J Vasc Med*. 2014;2014:160534. doi: 10.1155/2014/160534
 53. Quintana-Cabrera R, Fernandez-Fernandez S, Bobo-Jimenez V, Escobar J, Sastre J, Almeida A, Bolaños JP. γ -Glutamylcysteine detoxifies reactive oxygen species by acting as glutathione peroxidase-1 cofactor. *Nat Commun*. 2012;3:718. doi: 10.1038/ncomms1722
 54. Alshiek JA, Dayan L, Asleh R, Blum S, Levy AP, Jacob G. Anti-oxidative treatment with vitamin E improves peripheral vascular function in patients with diabetes mellitus and haptoglobin 2-2 genotype: a double-blinded cross-over study. *Diabetes Res Clin Pract*. 2017;131:200–207. doi: 10.1016/j.diabres.2017.06.026
 55. Breimer LH, Mikhailidis DP. Bilirubin and peripheral arterial disease: 15 years later. *Expert Opin Ther Targets*. 2009;13:139–140. doi: 10.1517/14728220802655321
 56. McDermott MM, Criqui MH, Domanchuk K, Ferrucci L, Guralnik JM, Kibbe MR, Kosmac K, Kramer CM, Leeuwenburgh C, Li L, et al. Cocoa to improve walking performance in older people with peripheral artery disease: the COCOA-PAD pilot randomized clinical trial. *Circ Res*. 2020;126:589–599. doi: 10.1161/CIRCRESAHA.119.315600
 57. Paapstel K, Kals J, Eha J, Tootsi K, Ottas A, Piir A, Jakobson M, Lieberg J, Zilmer M. Inverse relations of serum phosphatidylcholines and lysophosphatidylcholines with vascular damage and heart rate in patients with atherosclerosis. *Nutr Metab Cardiovasc Dis*. 2018;28:44–52. doi: 10.1016/j.numecd.2017.07.011
 58. Chen M, Miao G, Roman MJ, Devereux RB, Fabsitz RR, Zhang Y, Umans JG, Cole SA, Fiehn O, Zhao J. Longitudinal lipidomic profile of subclinical peripheral artery disease in American Indians: the Strong Heart Family Study. *Prev Chronic Dis*. 2025;22:E18. doi: 10.5888/pcd22.240220
 59. Huang F, Wang K, Shen J. Lipoprotein-associated phospholipase A2: the story continues. *Med Res Rev*. 2020;40:79–134. doi: 10.1002/med.21597
 60. Sui J, He M, Wang Y, Zhao X, He Y, Shi B. Sphingolipid metabolism in type 2 diabetes and associated cardiovascular complications. *Exp Ther Med*. 2019;18:3603–3614. doi: 10.3892/etm.2019.7981
 61. Harris KM, Mena-Hurtado C, Arham A, Burg MM, Freedland KE, Sinha R, Alabi O, Smolderen KG. Increasing prevalence of critical limb ischemia hospitalizations with distinct mental health burden among younger adults. *J Am Coll Cardiol*. 2021;78:2126–2128. doi: 10.1016/j.jacc.2021.09.025
 62. Meijer WT, Hoes AW, Rutgers D, Bots ML, Hofman A, Grobbee DE. Peripheral arterial disease in the elderly: the Rotterdam Study. *Arterioscler Thromb Vasc Biol*. 1998;18:185–192. doi: 10.1161/01.atv.18.2.185
 63. Xiao M, Zhong H, Xia L, Tao Y, Yin H. Pathophysiology of mitochondrial lipid oxidation: role of 4-hydroxynonenal (4-HNE) and other bioactive lipids in mitochondria. *Free Radic Biol Med*. 2017;111:316–327. doi: 10.1016/j.freeradbiomed.2017.04.363
 64. Liu H, Valdehuesa KNG, Ramos KRM, Nisola GM, Lee WK, Chung WJ. L-arabonate and D-galactonate production by expressing a versatile sugar dehydrogenase in metabolically engineered *Escherichia coli*. *Bioresour Technol*. 2014;159:455–459. doi: 10.1016/j.biortech.2014.03.056
 65. Huang HY, Appel LJ. Supplementation of diets with α -tocopherol reduces serum concentrations of γ - and δ -tocopherol in humans. *J Nutr*. 2003;133:3137–3140. doi: 10.1093/jn/133.10.3137
 66. Collins EG, Edwin Langbein W, Orebaugh C, Bammett C, Hanson K, Reda D, Edwards LC, Littoy FN. PoleStriding exercise and vitamin E for management of peripheral vascular disease. *Med Sci Sports Exerc*. 2003;35:384–393. doi: 10.1249/01.MSS.0000053658.82687.FF
 67. Törnwall ME, Virtamo J, Haukka J, Aro A, Albanes D, Edwards BK, Huttunen JK. Effect of α -tocopherol (vitamin E) and β -carotene supplementation on the

incidence of intermittent claudication in male smokers. *Arterioscler Thromb Vasc Biol*. 1997;17:3475–3480. doi: 10.1161/01.atv.17.12.3475

68. Törnwall ME, Virtamo J, Haukka JK, Aro A, Albanes D, Huttunen JK. The effect of alpha-tocopherol and beta-carotene supplementation on symptoms and progression of intermittent claudication in a controlled trial. *Atherosclerosis*. 1999;147:193–197. doi: 10.1016/s0021-9150(99)00176-8
69. Tang WW, Wang Z, Levison BS, Koeth RA, Britt EB, Fu X, Wu Y, Hazen SL. Intestinal microbial metabolism of phosphatidylcholine and cardiovascular risk. *N Engl J Med*. 2013;368:1575–1584. doi: 10.1056/NEJMoa1109400
70. Trøseid M, Ueland T, Hov J, Svardal A, Gregersen I, Dahl C, Aakhus S, Gude E, Bjørndal B, Halvorsen B. Microbiota-dependent metabolite trimethylamine-N-oxide is associated with disease severity and survival of patients with chronic heart failure. *J Intern Med*. 2015;277:717–726. doi: 10.1111/joim.12328
71. Kühn T, Rohrmann S, Sookthai D, Johnson T, Katzke V, Kaaks R, Von Eckardstein A, Müller D. Intra-individual variation of plasma trimethylamine-N-oxide (TMAO), betaine and choline over 1 year. *Clin Chem Lab Med*. 2017;55:261–268. doi: 10.1515/cclm-2016-0374
72. McEntyre CJ, Lever M, Chambers ST, George PM, Slow S, Elmslie JL, Florkowski CM, Lunt H, Krebs JD. Variation of betaine, N, N-dimethylglycine, choline, glycerophosphorylcholine, taurine and trimethylamine-N-oxide in the plasma and urine of overweight people with type 2 diabetes over a two-year period. *Ann Clin Biochem*. 2015;52:352–360. doi: 10.1177/0004563214545346
73. Floegel A, Drogan D, Wang-Sattler R, Prehn C, Illig T, Adamski J, Joost HG, Boeing H, Pischon T. Reliability of serum metabolite concentrations over a 4-month period using a targeted metabolomic approach. *PLoS One*. 2011;6:e21103. doi: 10.1371/journal.pone.0021103
74. Kühn T, Floegel A, Sookthai D, Johnson T, Rolle-Kampczyk U, Otto W, von Bergen M, Boeing H, Kaaks R. Higher plasma levels of lysophosphatidylcholine 18: 0 are related to a lower risk of common cancers in a prospective metabolomics study. *BMC Med*. 2016;14:13. doi: 10.1186/s12916-016-0552-3



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